Submission Guide: Property Capture Pipeline

Project: An end-to-end pipeline that processes consumer smartphone LiDAR captures into dimensioned 2D architectural floor plans and automated repair scopes of work. **Repository**: `https://github.com/Revanth-Kumar-Chilukuru/lidar-floorplan-pipeline`

Quick Start (3 Commands)

```
# 1. Clone and enter repo
git clone https://github.com/Revanth-Kumar-Chilukuru/lidar-floorplan-pipeline && cd lidar-floorpl

# 2. Setup environment (Python 3.11 required)
uv venv --python 3.11 && uv pip install -r requirements.txt

# 3. Reproduce all verified claims
uv run python scripts/reproduce_all.py
```

Device & Tier Matrix

Tier	Required Hardware	Status	What it Delivers
Tier 3 (LiDAR)	iPhone 12+ Pro, iPad Pro 2020+	**Active**	2D dimensioned floor plans, orthogonal wall footprint, metric scale, room boundaries.
Tier 2 (Video)	iPhone 15 or newer (per the case study)	**Not Implemented**	Explicitly fails cleanly. Requires GPU/SLAM.
Tier 1 (Photos)	Any modern smartphone	**Not Implemented**	Explicitly fails cleanly. Requires LayoutNet models.

Hardware Constraint Note: The author does not own an iPhone. All captures and code tests were run using the provided sample `Stray Scanner` dataset. As Stray Scanner requires an active LiDAR sensor, processing data from non-Pro phones or video/photo fallbacks is not currently possible on this CPU-bound pipeline without fabricating results.

Verification Status

Verified: - ✅ **Algorithmic Determinism**: Deterministic math replaces RANSAC/Open3D, ensuring 0 variance on the same data. - ✅ **Honest Null-Safety**: Missing ceilings or out-of-bounds constraints are strictly reported as "unmeasured" (null) rather than defaulted to arbitrary integers. - ✅ **Trajectory Bounding**: Extracted footprint fully encompasses the camera tracking path for all sample scans.

Not Verified: - ❌ **Physical Repeatability**: We cannot prove scan-to-scan physical reproducibility without multiple physical captures of the same room. - ❌ **Head-to-Head Accuracy**: No `Polycam` exports were available in the dataset for a true comparison. - ❌ **Opening Detection Accuracy**: "±2 cm on ≥85% gate" for opening detection remains unverified, as no laser/tape ground truth exists in the dataset.

Known Failure Modes

1. **Pose-Graph Drift**: True SLAM loop closure is not implemented. A PCA footprint alignment heuristic partially corrects orthogonal orientation, but cumulative odometry drift remains in large loops.
2. **Disconnected Wall Convex Hull**: To bound the trajectory, we rely on a Convex Hull which may overestimate concave or L-shaped rooms if large gaps exist in the walls. *Note: The single_room area varied between the full capture and the subset (33.4 m² vs 27.3 m²). Not fully explained; the footprint for this scan is sensitive to point density, so its area should not be trusted.*

Deliverables Index

- `README.md`: Entry point and reproduction steps.
- `TECHNICAL_REPORT.pdf`: Honest architectural analysis and algorithm decisions.
- `COMPLIANCE_MATRIX.md`: Mapping of prompt requirements to code realities.
- `FIX_LOOP_DECLARATION.md`: Before/after metrics for the determinism fix.
- `STOCK_CAPTURE_PROTOCOL.md`: Operator instructions for LiDAR capturing.
- `results/`: Output artifacts (JSON payloads, renders).